What Binds a Memory? Cache Slices, Recurrent States, and Why Separately Stored Facts Swap

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Abstract

A fact a user states in one conversation can be carried into the next by storing part of the model's own key/value cache and putting it back later. We ask which part carries the fact, and what goes wrong when several such memories are used together. On three frozen models (Qwen2.5-1.5B, the hybrid Qwen3.5-2B and Llama-3.2-3B), the answer reads a stated fact from the value's own position, late in the network, so writing a state at the subject cannot work. Two stored positions (the value and the full stop after it; on the hybrid, with its recurrent state) restore 0.87–1.00 clean use against 0.90–1.00 with the sentence in context, while the same two tokens as text give 0.24–0.27. On the hybrid, the two kinds of memory dissociate: the recurrent state recognises a name it is shown (yes/no 0.93) but cannot choose between two (0.48), and the attention entries choose (0.91) but cannot verify (0.50). Attention entries alone also refuse direct questions, and a generic "answer personal questions" direction removes most of that refusal. Slices stored in separate sessions swap partners when used together (partner confusion 0.33–0.41), because every one of them was encoded as the first fact of its conversation: storing a new memory as the second fact cuts confusion to 0.09–0.15, joint encoding writes the binding at the subject's positions, and storing the k-th memory as the k-th fact works for three memories but only halves the swaps for five, where neighbouring order tags blur.

1 Introduction

What has to be stored for a fact a user stated in one conversation to be *used* in a later one? Text retrieval answers by storing the words and re-reading them (Lewis et al., 2020; Packer et al., 2023). Weight editing stores an association in the MLPs (Meng et al., 2022, 2023). A third option keeps something from the model's own forward pass. Our earlier work stored the activation change an experience causes and added it back later as a steering vector (Jumle, 2026): the change carried leanings, but facts came out only as the word to say, almost never used in a sentence (at most 0.15 clean use against 0.96–0.98 in context). A pushed vector is one direction at one position, and in-context binding works through positions: the model binds an attribute to its entity with a shared order tag and fetches the attribute from its own position when asked (Feng and Steinhardt, 2024; Prakash et al., 2024, 2026; Dai et al., 2024).

This paper asks what a stored representation needs in order to be read by those circuits, using the model's own key/value cache as the store. Reinstating a cached turn is prompt caching (Gim et al., 2024); systems work reuses cached chunks for speed (Yao et al., 2025; Hu et al., 2025; Xiao et al., 2024). Our question is different: which part of the cache carries a fact, what a hybrid model's recurrent state adds, and why combining separately stored memories fails. Everything is training-free, on frozen models, with pre-registered decision rules (§2).

Contributions.

- **What carries a fact** (§3). The answer reads a stated value from the value's own position, late; the subject holds no copy. Two stored positions, the value and the full stop after it, restore the fact in a new session (0.87–1.00 clean use on three models), far beyond the same tokens as text (0.24–0.27).
- **What a hybrid's two memories do** (§4). On Qwen3.5-2B the recurrent state recognises a name it is shown and rejects look-alike names, but cannot choose between two names; the attention entries choose but cannot verify, and refuse direct questions. That refusal is mostly a generic response gate, not a memory-

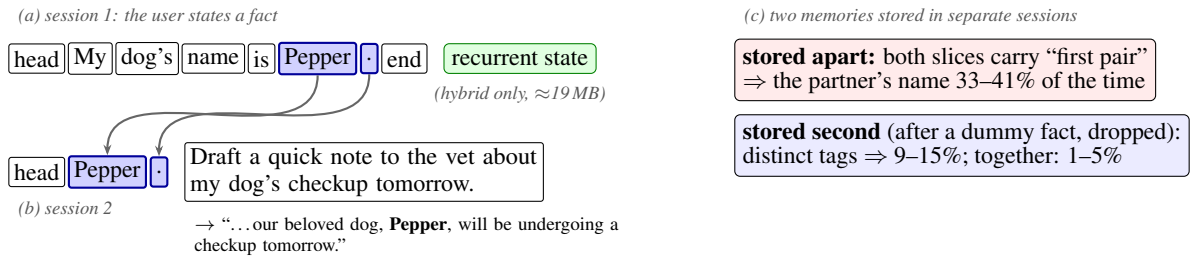


Figure 1: **(a–b)** A fact stated in session 1 is carried into session 2 by keeping the model's own keys and values at two positions, the value and the full stop after it (blue), and, on a hybrid model, the recurrent state after the turn; they are reinstated as a user-turn segment before a later question when its subject matches. **(c)** Slices stored in separate sessions were each encoded as the first fact of their conversation, and swap partners when used together; storing the new one as the second fact removes most swaps (§5).

specific signal.

- **Why separately stored facts swap, and a cheap fix** (§5). Every separately stored slice was encoded as the first fact of its conversation. Storing a new memory as a later fact removes most swaps without re-encoding; joint encoding writes the binding at the subject's positions; the fix works for three memories and halves the swaps for five.

2 Setup

Models. Qwen2.5-1.5B-Instruct (Qwen Team, 2024) and Llama-3.2-3B-Instruct (Grattafiori et al., 2024), both full attention, and Qwen3.5-2B (Qwen Team, 2026), a hybrid with full attention in 6 of its 24 blocks and Gated DeltaNet (Yang et al., 2025) in the other 18. Qwen3.5-9B is used teacher-forced only (Appendix B). All in fp32, frozen.

Facts. A memory is one sentence the user said, e.g. *"My dog's name is Pepper."* We use 7 hand-written facts (pet, family and work names) and 52 more drawn from a pool (30 people, 8 pets, 14 attributes such as a hometown). Each fact has *use* prompts (two per hand-written fact, one templated per pool fact) that need it without asking for it (*"Draft a quick note to the vet about my dog's checkup tomorrow"*), a *direct* question (*"What's my dog's name?"*) and balanced yes/no probes (*"Is my dog called Pepper?"* vs. an unrelated name).

Harvest and reinstate. Session 1 is the model's chat template with the sentence as the user turn. We keep its keys and values (K,V) at chosen positions, and on the hybrid the recurrent and convolution states after the turn. In session 2, the kept entries are placed as a segment after the template head and before the new user turn, with keys re-rotated to their new positions; the recurrent state, if kept, initialises the DeltaNet blocks. A gate puts the memory in only when the question's subject matches the stored subject string; with the gate, the answers to 50 unrelated prompts are token-identical to no memory (1.000 in every gated condition). Table 1 lists the conditions. Reinstating the whole turn reproduces the sentence as an earlier user turn exactly (max $|\Delta|$ logit $\leq 1.1\times10^{-4}$), which every run checks.

condition	what session 2 contains
CTX	the sentence in the user turn (ceiling)
NOMEM	nothing (floor)
full-turn slice	K,V of the whole stored turn
2-position slice	K,V at the value and the full stop
same tokens as text	"Pepper." as an earlier user turn
hybrid (Qwen3.5-2B): which parts of the memory	
attention entries	2 positions, 6 attention blocks
recurrent state	the DeltaNet state only, no K,V
both	2 positions + the state

Table 1: Conditions for one memory. Two-memory conditions are in §5.

Measures. Each prompt is answered once greedily and with four samples. *Clean use* means the answer uses the stored value correctly and is coherent, by the majority of three judges from different families (Qwen3.5-9B, gpt-oss and Gemma-4; a tie goes to the first). Teacher-forced measures read $\Delta \log P$ of the value at the answer prefix (*"Your dog's name is"*) and the yes/no margin at the first answer token. Intervals are 95% bootstrap CIs over items, then prompts (2000 resamples); differences are paired.

Pre-registration. The first experiments (where the value is read, the subject write, the slice conditions) were pre-registered with decisive rules; every

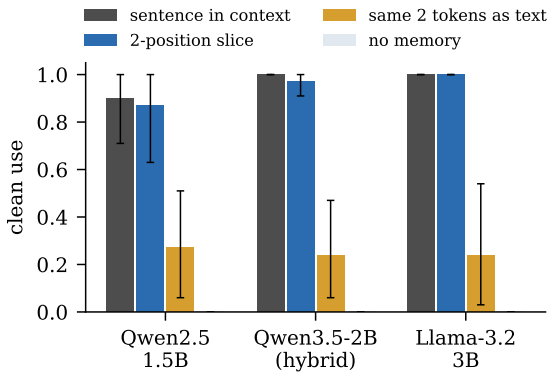


Figure 2: Clean use on the use prompts of the 7 hand-written facts. The 2-position slice (on the 2B with the recurrent state) is close to the sentence in context; the same two tokens as text are not.

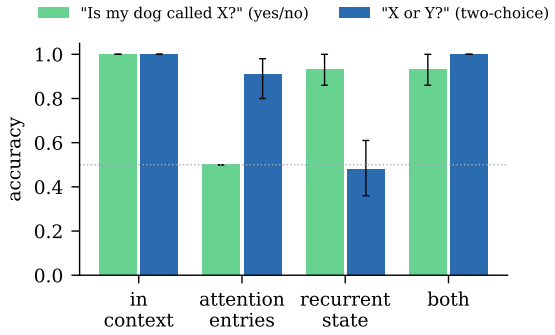


Figure 3: Qwen3.5-2B, 7 facts, teacher-forced. The recurrent state verifies a name it is shown but cannot choose between two; the attention entries choose but cannot verify. Dotted: chance.

later part was added as an addendum with its rule written and committed before any job ran. We report each rule's verdict, including the ones that fail (Appendix A). Code, rules and reports: `https://github.com/vedant-jumle/Seahorse`.

3 What carries a fact

The value is read from its own position. Patching the question's subject token between two facts moves at most 0.02 of the answer's preference, at every block and on all three models; patching the stored value moves all of it (1.00). Blocking the answer's attention to the value in Qwen2.5's last five blocks removes 0.59 of the gain; the rest reaches the answer through other positions, mainly the full stop. This is the in-context route (Prakash et al., 2026; Feng and Steinhardt, 2024) rather than enrichment at the subject (Geva et al., 2023), and it rules out a write at the subject: storing the subject's enriched state and adding it back gives 0.00 clean use.

Two positions are enough. Figure 2 shows clean use on the use prompts. Keeping only the value and the full stop gives 0.87 on Qwen2.5 (the sentence in context: 0.90), 1.00 on Llama (1.00) and, with the recurrent state, 0.97 on the 2B (1.00). A slice harvested after an unrelated exchange still works (0.80 on Qwen2.5, 1.00 on Llama), and on the 52 pool facts the 2-position slice reaches 0.97 (Qwen2.5) and 0.98 (Llama) on use prompts. Because the slice sits in a user-turn segment, the model treats the fact as the user's (on Qwen2.5 it never claims the fact as its own), unlike a steering vector (Jumle, 2026).

The slice carries more than its tokens. The same two tokens as an earlier user turn (*"Pepper."*) give 0.27, 0.24 and 0.24 (Figure 2; the slice minus the text $+0.60$ to $+0.76$, all CIs above 0). The stored K,V at "Pepper" and "." were computed with the whole sentence in view, so they carry its relation to the subject; as text, the model has to guess what "Pepper" refers to. The difference also holds for verification: on Llama, asked *"Is my dog called Pepper or Max?"*, the slice picks the right name 0.95 of the time and the text 0.68.

4 What a hybrid's two memories do

On Qwen3.5-2B the attention entries and the recurrent state can be kept separately. Teacher-forced, the 2-position attention entries alone carry the value ($\Delta \log P$ $+10.0$ nats; in context $+10.65$), the state alone only part of it ($+4.2$). In generation the attention entries alone give 0.67 clean use on use prompts but refuse direct questions (*"I don't know your dog's name! Could you tell me what it is?"*), and the state alone gives no recall at all (0.00).

The state recognises; the attention entries select. Two question forms separate the parts (Figure 3). Asked *"Is my dog called Pepper?"*, the state alone answers correctly 0.93 of the time (balanced against unrelated names), the attention entries alone 0.50 (always "No"). Asked *"Is my dog called Pepper or Max?"*, the order reverses: attention entries 0.91, state 0.48. The state signals that a name it is shown was said; the attention entries can pick out which name it was but not confirm it. The recognition sits in the recurrent state, not the convolution buffer (yes/no 0.86 vs. 0.50).

Recognition is of the item, and it does not bind. Afendulev et al. (2026) find, within one session on Qwen3.5, that the recurrent state supports recognition but also accepts associated lures, a gist familiarity. Across sessions, our state alone rejects lures more often than unrelated names: false "yes" 0.00 for look-alike names (Pepper/Piper) and 0.04 for same-theme names (Pepper/Ginger) against 0.07 for unrelated ones (balanced accuracy against lures 0.96 [0.88, 1.00]); on 38 pool facts, look-alike minus unrelated is -0.01 [-0.05, 0.03] (balanced 0.94). The pre-registered verdict is item recognition both times. But the joint state of two facts says yes to either name for either person (balanced 0.50 [0.40, 0.60]): it records which names were said, not whose they are.

Why the attention entries refuse. The value is there, so the refusal is a decision to answer. A difference-of-means "told" direction (the sentence in context minus no memory, at the last prompt token, built on other facts) restores direct answers from the attention entries (Table 2); a random direction of the same norm and the reversed sign do not, and the told direction without a memory makes the model invent a name rather than recall one. But a *comply* direction built with no memory at all, from personal questions the model refuses (*"What's my blood type?"*) minus matched general ones (*"What's the most common blood type?"*), does most of the same job (cosine with told 0.4–0.65 in the middle and late blocks). On the 2B the told direction beats it by only $+0.09$ [0.00, 0.18], so the pre-registered verdict is "generic comply"; on Qwen2.5 ($+0.09$ [0.03, 0.15]) and Llama ($+0.15$ [0.07, 0.25]) a small told-specific part remains. On held-out facts the told direction also lowers use-prompt clean use on the 2B and Qwen2.5 (-0.14, -0.18), which comply does not. The attention-only refusal is thus mostly the model's generic refusal of personal questions, close to the refusal direction of Arditi et al. (2024); the recurrent state lifts it in a way that is partly item-specific (own entries with another fact's state 0.65, with a placebo sentence's state 0.60, with its own 0.88).

5 Two or more memories

Separately stored slices swap partners. Two facts are stored in separate sessions and both slices are put back before a question about one of them (gates forced open). With 8 hand-written pairs, 5 of them confusable (two members of one kind, e.g. a first and a second dog) and 3 with different subjects, the answer gives the partner's name instead of the member's own in 0.37 (Qwen2.5), 0.33 (2B) and 0.41 (Llama) of confusable answers, against 0.01–0.05 when the two sentences were encoded together (Figure 4). Different subjects swap too (0.20–0.26). Placing the second slice after the first, rather than at overlapping positions, does not help, and on the 2B the swaps persist with the joint recurrent state, so they arise in the attention entries. On 34 pool pairs the same holds on all three models (0.16–0.22 vs. 0.00).

direct question, clean use	Qwen2.5	2B	Llama
attention-only slice (X)	0.72	0.12	0.62
X + told	0.89	0.76	0.88
X + told, random dir.	0.72	0.15	0.64
X + told, reversed	0.17	0.09	0.49
X + comply (no memory)	0.80	0.67	0.73
X + comply, random dir.	0.59	0.12	0.51
no memory + told	0.00	0.03	0.00
use prompts: $X \rightarrow X$+told	.98→.80	.68→.54	.99→.98

Table 2: Refusal of direct questions on 26 held-out pool facts (directions built on the other 26). X = the 2-position slice (on the 2B, attention entries only).

The cause is storage order. Binding IDs explain the swap if both slices carry the same ID (Feng and Steinhardt, 2024): each was encoded as the first and only fact of its conversation. We test this directly. The second member B is harvested after a dummy fact in the same turn (*"My gardener's name is Brian. My second dog's name is Rocky."*) and the dummy's tokens are dropped, so B's kept tokens are the same tokens as before (asserted) but were computed as the second fact. Confusion falls from 0.37/0.33/0.41 to 0.09/0.15/0.09, and on the pool pairs to 0.01 on every model, close to joint encoding. Two controls qualify this. An unrelated sentence before B (*"By the way, I had a coffee this morning."*) helps less (0.26/0.19/0.31; pool 0.07/0.05/0.13): the effect is not only that B is no longer at the start of its context, the reading of separately encoded chunks in position-independent caching (Hu et al., 2025; Liu et al., 2026b). And storing *both* members second, which should restore the shared tag, brings the swaps back only on Qwen2.5's pool pairs (0.20 vs. 0.22), and only partly elsewhere (0.03–0.21). The pre-registered rule, which required that, holds on Qwen2.5, on the 2B's hand-written pairs only, and not on Llama. Storage order explains most of the swaps; a smaller

part is generic.

Joint encoding writes the binding at the subject. Two stored positions per fact stay confused even when encoded together (0.20 on Qwen2.5, 0.17 on Llama), so the binding is not at the value. Adding the stored sentence's subject span (*"younger sister"*, *"friend from work"*) from the joint encoding gives joint encoding's result (0.03 and 0.01; pool pairs 0.00), while the same positions from separate encodings do not help (0.34, 0.29). The entity side of the pair's tag is written at the subject's positions, and only when the facts are read together. On the 2B the joint recurrent state already separates two-position slices (0.06).

The fix scales to three memories, not five. Storing the k-th memory of a set after $k-1$ dummy facts makes it the k-th fact (Figure 5). For three same-kind memories (eldest, middle and youngest sister; first to third dog) confusion falls from 0.47/0.38/0.41 to 0.09/0.04/0.09, against 0.00–0.01 for joint encoding; the pre-registered rule holds on Qwen2.5 and the 2B and misses on Llama by its overlap clause. For five same-kind pets it falls from 0.51–0.55 to 0.22–0.30 (joint 0.00–0.03), and the remaining errors are between neighbours: they concentrate at positions 3 to 5, and the fourth pet is answered with the fifth's or the third's name. Close binding IDs are confused (Feng and Steinhardt, 2024), and the IDs of later positions are closer. For memories about different subjects, storage order reaches joint encoding at both sizes on all three models (0.00–0.03). Placing the ordered slices in reverse order changes nothing (-0.02 to $+0.03$): what counts is the order a memory was stored in, not where its slice is put back.

6 Related work

Binding in context. Language models bind entities to attributes with binding IDs that are not tied to position, and confuse IDs that are close (Feng and Steinhardt, 2024; Dai et al., 2024); answers are fetched from the attribute's position through order-ID lookbacks (Prakash et al., 2024, 2026). Facts from the weights are instead enriched at the subject (Geva et al., 2023). We find the in-context route for facts said in an earlier session, show that separately cached facts share an ID, and locate the entity side at the subject's positions.

Cache reuse. Prompt Cache reuses separately encoded modules and notes that independence between them matters (Gim et al., 2024); CacheBlend, EPIC, Hypic and LinearKV recompute part of each reused chunk to recover the cross-chunk attention it lacks (Yao et al., 2025; Hu et al., 2025; Liu et al., 2026b,c); InfLLM retrieves cached blocks (Xiao et al., 2024); KV-State Grafting restores full cached blocks across sessions (Schelpe, 2026). These works measure speed and task accuracy. We measure which part of a cached fact carries it, why separately cached facts swap, and that a storage-time order fixes it for small sets without recompute.

Hybrid states. On hybrids, the attention cache carries retrieval and the recurrent state matters little for it (Yu et al., 2026); Afendulev et al. (2026) show recognition from the recurrent state within one session, as gist familiarity. We show item-specific recognition across sessions and its double dissociation from selection by the attention entries.

Trained and steered memories. Cartridges train a small KV cache per corpus (Eyuboglu et al., 2026); Memory Inception inserts text-derived KV at chosen layers for steering (Liu et al., 2026a); FwPKM trains a fast-weight memory (Zhao and Jones, 2026). Steering vectors as memory carry leanings but not facts (Jumle, 2026). Known-entity and refusal directions (Ferrando et al., 2025; Arditi et al., 2024) bear on why attention-only memories refuse.

7 Limitations

The models are small (1.5–3B in generation; 9B teacher-forced only), and the 2B is the only hybrid tested in generation. Memories are single sentences about names and simple attributes; we do not test long episodes, paraphrased questions beyond our prompts, or facts that must be combined. The two-memory results use oracle gates; retrieval by the subject's words works for 59 facts but turns a wrong pick into a confident wrong answer (Appendix C). Judges are models, checked against each other rather than against human labels. Most results come from addenda with rules fixed before the run, not from the first pre-registration, and several of their rules failed in part; we report those verdicts as they came out. The storage-order fix needs to know how many memories of a set exist, and fails for five same-kind memories, where joint encoding is still needed.

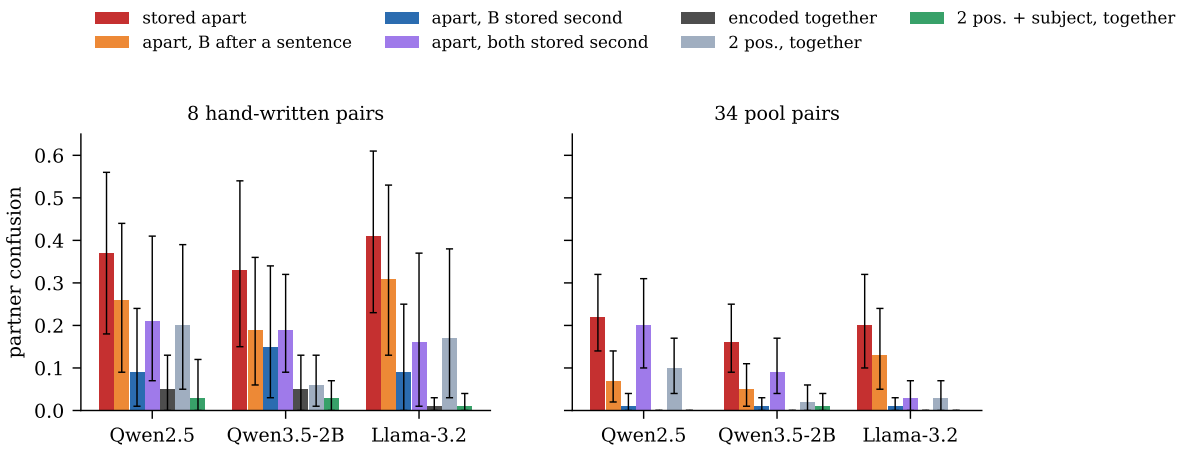


Figure 4: Partner confusion (the partner's value in the answer, the member's own not) for confusable pairs, gates forced open. Storing the second memory as the second fact (blue) removes most swaps; an unrelated sentence before it helps less; two positions encoded together stay confused on the full-attention models unless the subject's positions are kept too (green). On the 2B every condition uses the joint recurrent state.

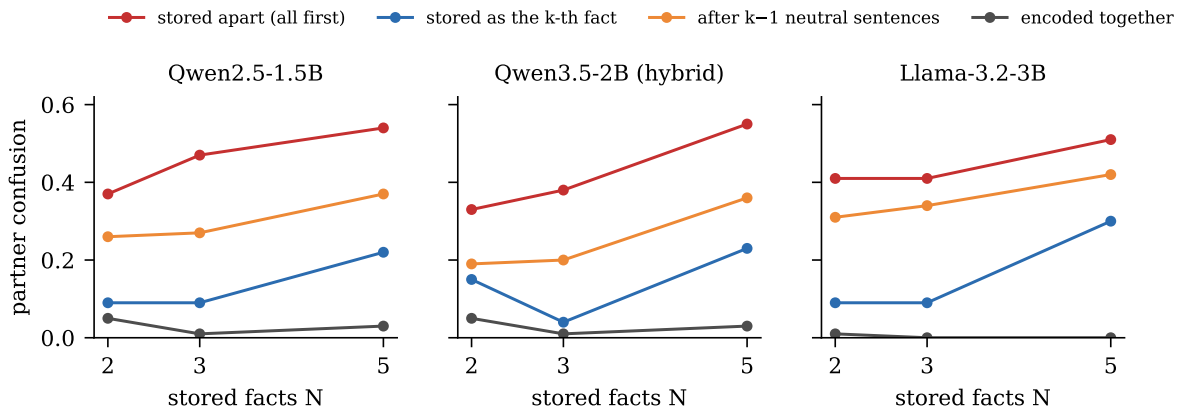


Figure 5: Partner confusion against the number of same-kind memories used together (N = 2: the hand-written pairs; N = 3, 5: 12 and 8 groups). Storing the k-th memory as the k-th fact tracks joint encoding up to three and diverges at five.

8 Conclusion

A stated fact lives at its own position, and a two-position slice of the model's own cache carries it into a new conversation better than the same words do. On a hybrid, recognition and selection are split between the recurrent state and the attention entries. The main failure, swaps between separately stored memories, comes from storage order: every memory thinks it is the first fact. Giving each memory a different place in a stored sequence, or re-reading the memories that are used together, is work that an offline consolidation phase can do and plain retrieval over text does not.

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A Pre-registered rules and verdicts

Table 3 lists every decision rule behind a claim in the paper, with its verdict. A judge-pipeline bug left some answers with only the first judge's label (identical answers across models were judged once and keyed to the first model); after the fix every answer has all three labels, and no verdict changed (clean use moved by at most 0.02).

B Qwen3.5-9B, teacher-forced

On the 9B the attention entries alone carry the value ($+9.83$ nats; in context $+10.80$) with yes/no at chance, as on the 2B. The state alone never moves the true name's yes/no margin above zero: the 9B says "No" to anything it does not know (margin about -3.9), and the state raises the true name's margin to -1.27 against -3.15 for unrelated names and -3.2 to -3.3 for lures. The margins show item-specific recognition (about $+2.6$ nats on the true name, $+0.7$ on foils) too weak to overturn the prior, so the binary rule fails.

C Retrieval and the binding-ID repair

With slices keyed on the subject's words, the right slice is found for 0.85–0.92 of direct questions at 10, 20 and 59 stored facts on Qwen2.5 (a whole-text key: 0.42–0.61), and the gate fires on unrelated prompts 0–4% of the time. A wrong pick is worse than none: the model states the wrong fact with confidence. Separately, a difference-of-means binding-ID direction is real in the late blocks (it flips which partner the model prefers in 0.75 of held-out contexts, a random direction 0), but adding it to one stored slice does not reduce swaps in generation (-0.02 [-0.14, 0.11]).

D Example answers

- 2-position slice, Qwen2.5, *"Draft a quick note to the vet about my dog's checkup tomorrow"*: *"Subject: Checkup for Pepper, Our Family's Pet. Dear [Vet's Name], I hope this message finds you well. I am writing to inform you that our beloved dog, Pepper, will be undergoing a checkup tomorrow…"*
- Attention entries only, 2B, *"Do you remember my sister's name?"*: *"I don't have access to personal information about your family members, including your sister's name. As an AI, I don't have the ability to remember past conversations…"*
- Attention entries + told, 2B: *"Your ferret's name is **Shadow**."*; + comply: *"The name of your coworker is **Chloe**."*
- No memory + told, 2B: *"Your dentist's name is **Dr. Sarah Jenkins**."* (invented).
- Five stored pets, the fourth asked: *"Your fourth cat's name is Atlas."* (Atlas is the fifth cat).

rule	models	verdict	where
Value read at the subject (stage 0, patch + knockout)	Qwen2.5, 2B	pointer-suspected (not at subject)	initial
Subject write gives clean use	Qwen2.5	fails (0.00)	initial
Reinstatement wins over the subject write	Qwen2.5	holds	arm R
Slice $>$ text (2-position slice $-$ same tokens, CI > 0)	all three	holds	add. 5, 7
Separately stored facts collide (sep $-$ joint confusion, CI > 0)	all three	holds (8 and 34 pairs)	add. 5, 7, 8
Cheap repair by the binding-ID direction	Qwen2.5	fails	add. 6
State recognition is item recognition, not gist	2B	holds (7 and 38 facts)	add. 7, 8
Missing told signal	2B / Qwen2.5 / Llama	holds (add. 7); held out: fails / fails / holds	add. 7, 8
Told is specific vs. comply (held out)	2B / Qwen2.5 / Llama	generic comply / specific / specific	add. 8
Pool gap is a prompt-type mismatch	Qwen2.5, Llama	holds	add. 7
The 2B value/verification split replicates on the 9B	9B	fails (strong "No" prior)	add. 7
Cheap re-encode of the new memory only (2 positions)	all three	fails	add. 7
The tag causes the swaps (stored second; both second $\approx$ apart)	Qwen2.5 / 2B / Llama	holds / partly / fails	add. 8
The entity side carries the tag (2 positions + subject)	Qwen2.5, Llama	holds	add. 8
Two-choice verification usable on Llama	Llama	holds	add. 8
Storage order scales, confusable, N = 3	Qwen2.5 / 2B / Llama	holds / holds / fails (overlap)	add. 9
Storage order scales, confusable, N = 5	all three	fails	add. 9

Table 3: Decision rules, in the order they were committed. "add." = an addendum to the pre-registration, written and committed before its jobs ran. One verdict printed by our report code was corrected by hand: the 2B's told vs. comply lower bound was 4×10^{-18}, printed as "> 0"; by the rule's order the verdict is "generic comply". Later rules treat $|\text{bound}| < 10^{-9}$ as 0.